

f-plane

Na dinâmica dos fluidos geofísicos, a aproximação do f-plane é uma aproximação onde o parâmetro de Coriolis, denotado f , é ajustado para um valor constante.

Essa aproximação é freqüentemente usada para a análise de ciclones tropicais altamente idealizados. O uso de um parâmetro constante de Coriolis evita a formação de beta-giros que são amplamente responsáveis pela direção norte-oeste da maioria dos ciclones tropicais.

Plano beta $f=f_0 + \beta y$

Na dinâmica dos fluidos geofísicos, uma aproximação pela qual o parâmetro de Coriolis, f , é ajustado para variar linearmente no espaço é chamado de aproximação do plano beta.

Em uma esfera rotativa como a Terra, varia com o seno de latitude; na assim chamada aproximação f-plane, essa variação é ignorada, e um valor de f apropriado para uma determinada latitude é usado em todo o domínio.

3- Ondas de Gravidade Inercial e distribuição de Variáveis

Nesta seção nos discutiremos o efeito da diferença centrada no espaço sobre as ondas de gravidade. Assim, nos consideramos o sistema de equação linearizada.

$$\frac{\partial u}{\partial t} = -g \frac{\partial h}{\partial x} + f v \quad 3.1a$$

$$\frac{\partial v}{\partial t} = -g \frac{\partial h}{\partial x} - f u \quad 3.1b$$

$$\frac{\partial h}{\partial t} = -H \nabla \cdot \vec{V} \quad 3.1c$$

Esta equação difere daquela da seção 2 no termo de coriolis f . O termo de coriolis não contém derivadas. Entretanto, eles são difíceis de calcular sobre a grade C, que foi ideal para ondas de gravidade puras.

Assim, nos reconsideramos o problema da distribuição de variáveis.

Não é óbvio como nós podemos analisar vários arranjos de variáveis. Nossa primeira opção é considerar (eq. 3.1) como parte de um sistema completo de equações primitivas. Nós estamos interessados no movimento de grande escala, por outro lado nos não devemos incluir o termo de Coriolis.

Sobre a grande escala, a equação primitiva admite dois tipos movimento distintos: baixa frequência e quase-geostrófico e escoamento quase-não divergente; e alta frequência ondas de

gravidade inercial. Ondas de gravidade inercial são continuamente excitadas na atmosfera, entretanto, como elas são dispersivas, uma acumulação local de energia de ondas dispersa com o tempo. Estes processos são conhecidos como ajustamento geostrófico; o movimento permanente é um balanço aproximadamente geostrófico e muda somente lentamente com o tempo. Neste capítulo nos estaremos concentrando numa simulação correta destes processos, em que é essencialmente governado pela equação de ondas de gravidade inercial(3.1).

Nós estamos interessados em ambas ondas causadas pelo efeito físico, e que é causado por inadequados dados iniciais e procedimento numérico.

Entretanto os detalhes do processo de ajustamento não importam tanto quanto a correção do resultado do escoamento quase-geostrófico.

Nós devemos, no entanto, investigar o efeito da distribuição do espaço de variáveis dependentes sobre a propriedade dispersiva das ondas de gravidade inercial. Este será feito usando a mais simples aproximação centrada para a derivada no espaço deixando a derivada no tempo em sua forma diferencial.

A discussão é baseada sobre aquilo que Winninghoff e Arakawa como apresentado por Arakawa (Arakawa, 1972; Arakawa et al. 1974).

Nos consideramos 5 caminhos de distribuição de variáveis dependentes. No espaço.

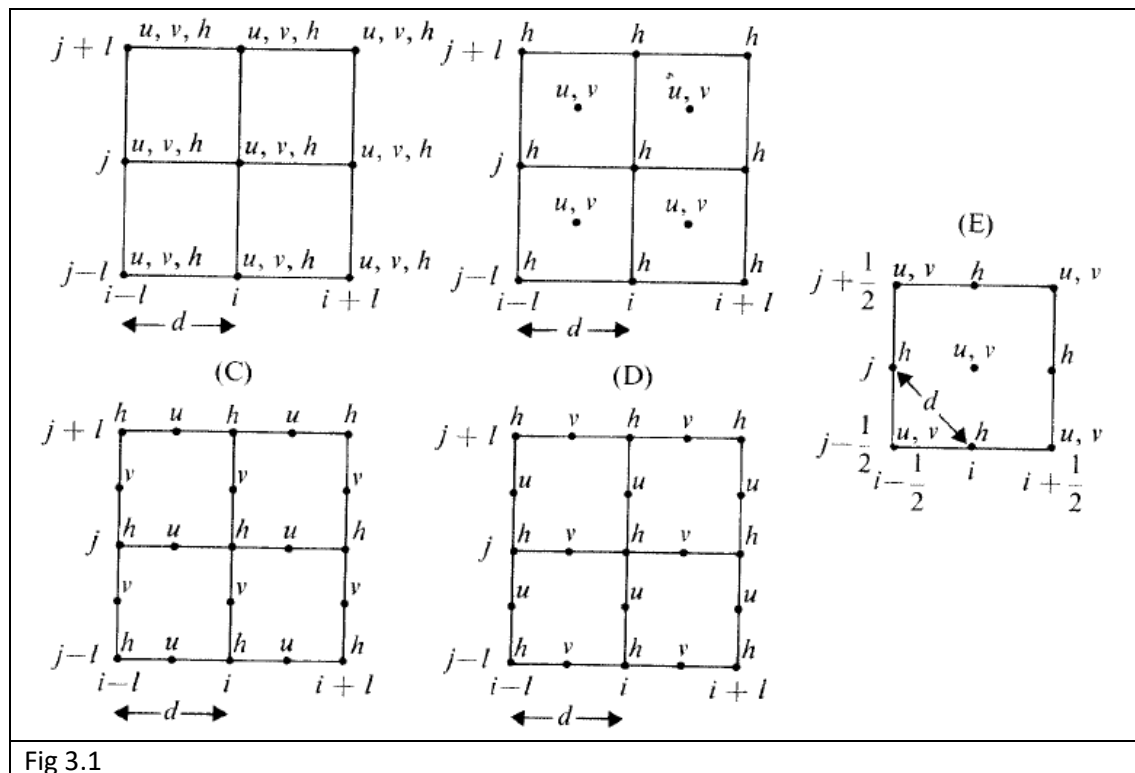


Fig 3.1

Nos Definimos d a distancia mais curta entre os pontos vizinhos carregando a mesma variável dependente. Na figura 3.1 d é o mesmo para cada uma das cinco grades. Assim, todas as grades tem o mesmo número de variáveis dependentes por unidade de área. A tempo de computação necessário para uma integração sobre cada uma das grades será sobre a mesma; propriedade

da solução obtida embora, será diferente devido ao efeito do espaço de arranjo das variáveis.

Usando os subscritos mostrado na figura 3.1, nos definimos um operador para a diferenciação no espaço centrado.

$$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$$

Esta rotação é aplicável a todas as grades. Aqui d' é a distancia entre os pontos os quais a diferença finita é feita. Assim, para a grade A, embora D, d' é igual ao tamanho da grade d, e para a grade E é igual a $\sqrt{2}d$.

Nos também definimos uma media sobre o mesmo dois pontos por:

$$(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$$

Assim, $(\delta_y \alpha)_{i,j}$ e $(\bar{\alpha}^y)_{i,j}$ são definido no mesmo caminho, mas com respeito ao eixo y.

Finalmente,

$$(\bar{\alpha}^{xy})_{i,j} \equiv (\bar{\bar{\alpha}^x}^y)_{i,j}$$

Para cada uma das 5 grades nos usamos uma aproximação centrada simples para a derivada no espaço e temos de coriolis (3.1). Obtemo-nos os diferentes sistemas:

GRADE A

$$\begin{aligned} \frac{\partial u}{\partial t} &= -g \bar{\delta}_x \bar{h}^x + f v = -g \left(\frac{1}{d'} \left(\bar{h}_{i+\frac{1}{2},j}^x - \bar{h}_{i-\frac{1}{2},j}^x \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(h_{i+\frac{1}{2}+\frac{1}{2},j} + h_{i+\frac{1}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(h_{i-\frac{1}{2}+\frac{1}{2},j} + h_{i-\frac{1}{2}-\frac{1}{2},j} \right) \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} (h_{i+1,j} + h_{i,j}) - \frac{1}{2} (h_{i,j} + h_{i-1,j}) \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} (h_{i+1,j} + h_{i,j} - h_{i,j} - h_{i-1,j}) \right) \right) + f v \\ &= -g \left(\frac{1}{2d'} (h_{i+1,j} - h_{i-1,j}) \right) + f v \end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\overline{\delta_y h^y} - fu = -g\left(\frac{1}{d'}\left(\bar{h}_{i,j+\frac{1}{2}}^y - \bar{h}_{i,j-\frac{1}{2}}^y\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i,j+\frac{1}{2}+\frac{1}{2}} + h_{i,j+\frac{1}{2}-\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i,j-\frac{1}{2}+\frac{1}{2}} + h_{i,j-\frac{1}{2}-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(h_{i,j+1} + h_{i,j}) - \frac{1}{2}(h_{i,j} + h_{i,j-1})\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(h_{i,j+1} + h_{i,j} - h_{i,j} - h_{i,j-1})\right)\right) - fu \\
&= -g\left(\frac{1}{2d'}(h_{i,j+1} - h_{i,j-1})\right) - fu
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x u^x} + \overline{\delta_y v^y}) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^x - \bar{h}_{i-\frac{1}{2},j}^x\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^y - \bar{v}_{i,j-\frac{1}{2}}^y\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(u_{i+\frac{1}{2}+\frac{1}{2},j} + u_{i+\frac{1}{2}-\frac{1}{2},j}\right) - \frac{1}{2}\left(u_{i-\frac{1}{2}+\frac{1}{2},j} + u_{i-\frac{1}{2}-\frac{1}{2},j}\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(v_{i,j+\frac{1}{2}+\frac{1}{2}} + v_{i,j+\frac{1}{2}-\frac{1}{2}}\right) - \frac{1}{2}\left(v_{i,j-\frac{1}{2}+\frac{1}{2}} + v_{i,j-\frac{1}{2}-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}(u_{i+1,j} + u_{i,j}) - \frac{1}{2}(u_{i,j} + u_{i-1,j})\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}(v_{i,j+1} + v_{i,j}) - \frac{1}{2}(v_{i,j} + v_{i,j-1})\right)\right) \\
&= H\left(\frac{1}{2d'}(u_{i+1,j} - u_{i-1,j})\right) + H\left(\frac{1}{2d'}(v_{i,j+1} - v_{i,j-1})\right)
\end{aligned}$$

GRADE B

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g\overline{\delta_x h^y} + fv = -g\left(\frac{1}{d'}\left(\bar{h}_{i+\frac{1}{2},j}^y - \bar{h}_{i-\frac{1}{2},j}^y\right)\right) + fv \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) + fv \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) + fv \\
&= -g\left(\frac{1}{2d'}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right) + fv
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\overline{\delta_y h^x} - fu = -g\left(\frac{1}{d'}\left(\bar{h}_{i,j+\frac{1}{2}}^x - \bar{h}_{i,j-\frac{1}{2}}^x\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} + h_{i-\frac{1}{2},j+\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i+\frac{1}{2},j-\frac{1}{2}} + h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{2d'}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right) - fu
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x u^y} + \overline{\delta_y v^x}) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^y - \bar{h}_{i-\frac{1}{2},j}^y\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^x - \bar{v}_{i,j-\frac{1}{2}}^x\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}}\right) - \frac{1}{2}\left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}}\right) - \frac{1}{2}\left(v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\left(u_{i+\frac{1}{2},j+\frac{1}{2}} - u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} - u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{2d'}\left(\left(v_{i+\frac{1}{2},j+\frac{1}{2}} - v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} - v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

GRADE C

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g\delta_x h + f\bar{v}^{xy} = -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) + f\left(\frac{1}{2}\left(\bar{v}_{i+\frac{1}{2},j}^y + \bar{v}_{i-\frac{1}{2},j}^y\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) \\
&\quad + f\left(\frac{1}{2}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}}\right) + \frac{1}{2}\left(v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) \\
&\quad + f\left(\left(\frac{1}{4}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\delta_y h - f\bar{u}^{xy} = -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) - f\left(\frac{1}{2}\left(\bar{u}^y_{i+\frac{1}{2},j} + \bar{u}^y_{i-\frac{1}{2},j}\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) \\
&\quad - f\left(\frac{1}{2}\left(\frac{1}{2}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}}\right) + \frac{1}{2}\left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\delta_x u + \delta_y v) = H\left(\left(\frac{1}{d'}\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j}\right)\right) + \left(\frac{1}{d'}\left(v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\frac{\partial u}{\partial t} = -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) + f\left(\left(\frac{1}{4}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)$$

$$\frac{\partial v}{\partial t} = -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)$$

$$\frac{\partial h}{\partial t} = H\left(\frac{1}{d'}\left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right)$$

GRADE D

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g \overline{\delta_x h^{xy}} + f \bar{v}^{xy} = -g \left(\frac{1}{d'} \left(\bar{h}_{i+\frac{1}{2},j}^{xy} - \bar{h}_{i-\frac{1}{2},j}^{xy} \right) \right) + f \left(\frac{1}{2} \left(\bar{v}_{i+\frac{1}{2},j}^y + \bar{v}_{i-\frac{1}{2},j}^y \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+\frac{1}{2},j+\frac{1}{2}}^y + \bar{h}_{i+\frac{1}{2},j-\frac{1}{2}}^y \right) - \frac{1}{2} \left(\bar{h}_{i-\frac{1}{2},j+\frac{1}{2}}^y + \bar{h}_{i-\frac{1}{2},j-\frac{1}{2}}^y \right) \right) \right) \\
&\quad + f \left(\frac{1}{2} \left(\frac{1}{2} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} \right) + \frac{1}{2} \left(v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+1,j}^y + \bar{h}_{i,j}^y \right) - \frac{1}{2} \left(\bar{h}_{i,j}^y + \bar{h}_{i-1,j}^y \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) = \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+1,j}^y - \bar{h}_{i-1,j}^y \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\frac{1}{2} \left(h_{i+1,j+\frac{1}{2}} + h_{i+1,j-\frac{1}{2}} \right) - \frac{1}{2} \left(h_{i-1,j+\frac{1}{2}} + h_{i-1,j-\frac{1}{2}} \right) \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{4d'} \left(\left(h_{i+1,j+\frac{1}{2}} - h_{i-1,j+\frac{1}{2}} + h_{i+1,j-\frac{1}{2}} - h_{i-1,j-\frac{1}{2}} \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g \overline{\delta_y h^{xy}} - f \bar{u}^{xy} = -g \left(\frac{1}{d'} \left(\bar{h}_{i,j+\frac{1}{2}}^{xy} - \bar{h}_{i,j-\frac{1}{2}}^{xy} \right) \right) - f \left(\frac{1}{2} \left(\bar{u}_{i+\frac{1}{2},j}^y + \bar{u}_{i-\frac{1}{2},j}^y \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i,j+\frac{1}{2}+\frac{1}{2}}^x + \bar{h}_{i,j+\frac{1}{2}-\frac{1}{2}}^x \right) - \frac{1}{2} \left(\bar{h}_{i,j-\frac{1}{2}+\frac{1}{2}}^x + \bar{h}_{i,j-\frac{1}{2}-\frac{1}{2}}^x \right) \right) \right) \\
&\quad - f \left(\frac{1}{2} \left(\frac{1}{2} \left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} \right) + \frac{1}{2} \left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} (\bar{h}_{i,j+1}^x + \bar{h}_{i,j}^x) - \frac{1}{2} (\bar{h}_{i,j}^x + \bar{h}_{i,j-1}^x) \right) \right) \\
&\quad - f \left(\left(\frac{1}{4} \left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) = \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} (\bar{h}_{i,j+1}^y - \bar{h}_{i,j-1}^y) \right) \right) \\
&\quad - f \left(\left(\frac{1}{4} \left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\frac{1}{2} (h_{i+\frac{1}{2},j+1} + h_{i-\frac{1}{2},j+1}) - \frac{1}{2} (h_{i+\frac{1}{2},j-1} + h_{i-\frac{1}{2},j-1}) \right) \right) \right) \\
&\quad - f \left(\left(\frac{1}{4} \left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{4d'} \left((h_{i+\frac{1}{2},j+1} - h_{i+\frac{1}{2},j-1} + h_{i-\frac{1}{2},j+1} - h_{i-\frac{1}{2},j-1}) \right) \right) \\
&\quad - f \left(\left(\frac{1}{4} \left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x u^{xy}} + \overline{\delta_y v^{xy}}) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^{xy} - \bar{u}_{i-\frac{1}{2},j}^{xy}\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^{xy} - \bar{v}_{i,j-\frac{1}{2}}^{xy}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{u}_{i+\frac{1}{2}+\frac{1}{2},j}^y + \bar{u}_{i+\frac{1}{2}-\frac{1}{2},j}^y\right) - \frac{1}{2}\left(\bar{u}_{i-\frac{1}{2}+\frac{1}{2},j}^y + \bar{u}_{i-\frac{1}{2}-\frac{1}{2},j}^y\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{v}_{i,j+\frac{1}{2}+\frac{1}{2}}^x + \bar{v}_{i,j+\frac{1}{2}-\frac{1}{2}}^x\right) - \frac{1}{2}\left(\bar{v}_{i,j-\frac{1}{2}+\frac{1}{2}}^x + \bar{v}_{i,j-\frac{1}{2}-\frac{1}{2}}^x\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{u}_{i+1,j}^y + \bar{u}_{i,j}^y\right) - \frac{1}{2}\left(\bar{u}_{i,j}^y + \bar{u}_{i-1,j}^y\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{v}_{i,j+1}^x + \bar{v}_{i,j}^x\right) - \frac{1}{2}\left(\bar{v}_{i,j}^x + \bar{v}_{i,j-1}^x\right)\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\left(\bar{u}_{i+1,j}^y - \bar{u}_{i-1,j}^y\right)\right)\right) + H\left(\frac{1}{2d'}\left(\left(\bar{v}_{i,j+1}^x - \bar{v}_{i,j-1}^x\right)\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\frac{1}{2}\left(u_{i+1,j+\frac{1}{2}} + u_{i+1,j-\frac{1}{2}}\right) - \frac{1}{2}\left(u_{i-1,j+\frac{1}{2}} + u_{i-1,j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{2d'}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+1} + v_{i-\frac{1}{2},j+1}\right) - \frac{1}{2}\left(v_{i+\frac{1}{2},j-1} + v_{i-\frac{1}{2},j-1}\right)\right)\right) \\
&= H\left(\frac{1}{4d'}\left(\left(u_{i+1,j+\frac{1}{2}} + u_{i+1,j-\frac{1}{2}}\right) - \left(u_{i-1,j+\frac{1}{2}} + u_{i-1,j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{4d'}\left(\left(v_{i+\frac{1}{2},j+1} + v_{i-\frac{1}{2},j+1}\right) - \left(v_{i+\frac{1}{2},j-1} + v_{i-\frac{1}{2},j-1}\right)\right)\right)
\end{aligned}$$

$$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$$

$$(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$$

$$\frac{\partial u}{\partial t} = -g \delta_x h + f v = -g \left(\frac{1}{d'} \left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j} \right) \right) + f v_{i,j}$$

$$\frac{\partial v}{\partial t} = -g \delta_y h - f u = -g \left(\frac{1}{d'} \left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}} \right) \right) - f u_{i,j}$$

$$\begin{aligned} \frac{\partial h}{\partial t} &= -H(\delta_x u + \delta_y v) = H \left(\left(\frac{1}{d'} \left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} \right) \right) + \left(\frac{1}{d'} \left(v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}} \right) \right) \right) \\ &= H \left(\frac{1}{d'} \left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}} \right) \right) \right) \end{aligned}$$

Nos podemos primeiro analisar um caso unidimensional, que na qual as variáveis u, v, h não variam com y , assim nos temos

$$u, v, h = u(x, t), v(x, t), h(x, t)$$

A equação 3.1 se reduz a

$\frac{\partial u}{\partial t} = -g \frac{\partial h}{\partial x} + f v$ $\frac{\partial v}{\partial t} = -f u$ $\frac{\partial h}{\partial t} = -H \frac{\partial u}{\partial x}$	3.3
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Substituindo a 1,2

$$u(x, t) = \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial t} = -iv \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (-iv)(-iv)\text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -v^2 \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$h(x, t) = \text{Re}[\hat{h}e^{-i(kx-vt)}]$$

$\frac{\partial^2 u}{\partial^2 t} = -g \frac{\partial}{\partial x} \frac{\partial h}{\partial t} + f \frac{\partial v}{\partial t}$ $\frac{\partial v}{\partial t} = -fu$ $\frac{\partial h}{\partial t} = -H \frac{\partial u}{\partial x}$	3.3
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$\frac{\partial^2 u}{\partial^2 t} = -g \frac{\partial}{\partial x} \left(-H \frac{\partial u}{\partial x} \right) + f(-fu)$	3.3
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$\frac{\partial^2 u}{\partial^2 t} = gH \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) - f^2 u$	3.3
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$$u(x, t) = \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial t} = iv \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (iv)(iv)\text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -v^2 \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial x} = -ik \text{Re}[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 x} = (-ik)(-ik)Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 x} = -k^2 Re[\hat{u}e^{-i(kx-vt)}]$$

$-v^2 Re[\hat{u}e^{-i(kx-vt)}] = -gHk^2 Re[\hat{u}e^{-i(kx-vt)}] - f^2 Re[\hat{u}e^{-i(kx-vt)}]$	3.3
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$-v^2 = -gHk^2 - f^2$	3.3
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$v^2 = gHk^2 + f^2$	3.3
$\left(\frac{v}{f}\right)^2 = \frac{gH}{f^2} k^2 + 1$	3.3

$\left(\frac{v}{f}\right)^2 = \frac{\lambda^2}{f} k^2 + 1$	3.3
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Assim, o raio de deformação:

$$\lambda = \frac{\sqrt{gH}}{f}$$

Nunca é igual a zero, a frequência da onda de gravidade inercial é uma função de k monotonicamente crescente. Portanto a velocidade de grupo $\frac{\partial v}{\partial k}$ nunca será igual a zero.

Isto é muito importante para os processos de ajustamento geotrópico. pois impede uma acumulação local de energia das ondas.

No focaremos no efeito de diferenciação finita no espaço. As variaveis são assumidas não dependente de y, o sistema 3.2 reduz a:

$$\frac{\partial u}{\partial t} = -g\overline{\delta_x h^x} + fv$$

$$\frac{\partial v}{\partial t} = -fu$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})$$

$$\frac{\partial u}{\partial t} = -g\overline{\delta_x h^x} + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{d'}\left(\bar{h}^x_{i+\frac{1}{2},j} - \bar{h}^x_{i-\frac{1}{2},j}\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2}+\frac{1}{2},j} + h_{i+\frac{1}{2}-\frac{1}{2},j}\right) - \frac{1}{2}\left(h_{i-\frac{1}{2}+\frac{1}{2},j} + h_{i-\frac{1}{2}-\frac{1}{2},j}\right)\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{2d'}\left((h_{i+1,j} + h_{i,j}) - (h_{i,j} + h_{i-1,j})\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{2d'}\left((h_{i+1,j} - h_{i-1,j})\right) + fv$$

$$\frac{\partial^2 u}{\partial^2 t} = -g\frac{1}{2d'}\left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t}\right)\right) + fv$$

$$\frac{\partial^2 u}{\partial^2 t} = -g\frac{1}{2d'}\left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t}\right)\right) + fv$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{d'} \left(\bar{u}^x_{i+1+\frac{1}{2},j} - \bar{u}^x_{i+1-\frac{1}{2},j} \right) = -H \frac{1}{d'} \left(\bar{u}^x_{i+\frac{3}{2},j} - \bar{u}^x_{i+\frac{1}{2},j} \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{d'} \left(\frac{1}{2} \left(u^x_{i+\frac{3}{2}+\frac{1}{2},j} + u^x_{i+\frac{3}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(u^x_{i+\frac{1}{2}+\frac{1}{2},j} + u^x_{i+\frac{1}{2}-\frac{1}{2},j} \right) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} + u^x_{i+1,j}) - (u^x_{i+1,j} + u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\bar{u}^x_{i-1+\frac{1}{2},j} - \bar{u}^x_{i-1-\frac{1}{2},j} \right) = -H \frac{1}{d'} \left(\bar{u}^x_{i-\frac{1}{2},j} - \bar{u}^x_{i-\frac{3}{2},j} \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\frac{1}{2} \left(u^x_{i-\frac{1}{2}+\frac{1}{2},j} + u^x_{i-\frac{1}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(u^x_{i-\frac{3}{2}+\frac{1}{2},j} + u^x_{i-\frac{3}{2}-\frac{1}{2},j} \right) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\frac{1}{2} (u^x_{i,j} + u^x_{i-1,j}) - \frac{1}{2} (u^x_{i-1,j} + u^x_{i-2,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right)$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t} \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(-H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right) - \left(-H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(-H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right) + H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = \frac{gH}{4d'^2} \left(\left(\left((u^x_{i+2,j} - u^x_{i,j}) \right) - \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = \frac{gH}{4d'^2} \left(\left((u^x_{i+2,j} + u^x_{i-2,j}) - (2u^x_{i,j}) \right) \right) + f v$$

$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$	
$(\delta_x \delta_x \alpha)_{i,j} \equiv \frac{1}{d'^2} \left(\alpha_{i+1,j} + \alpha_{i-1,j} - 2\alpha_{i,j} \right)$	
$(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$	

$$u(x, t) = Re[\hat{u}e^{-i(kx-vt)}]$$

$$h(x, t) = Re[\hat{h}e^{-i(kx-vt)}]$$

$$x = j\Delta x$$

$$t = m\Delta t$$

$$u(x, t) = Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$h(x, t) = Re[\hat{h}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial u}{\partial t} = -i(-v)Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -i(-mv)(-i(-mv))Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (imv)(imv)Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -(mv)^2 Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u(j = n, m = 1)}{\partial^2 t} = -(v)^2 Re[\hat{u}e^{-i(k(n)\Delta x-v\Delta t)}]$$

$$\frac{\partial^2 u(j = n, m = 1)}{\partial^2 t} = -(v)^2 Re[\hat{u}(e^{-i(nk\Delta x-v\Delta t)})]$$

$$u(j = n, m = 1) = Re[\hat{u}e^{-i(kn\Delta x-v(\Delta t))}]$$

$$\begin{aligned} u(j = n + 2, m = 1) &= Re[\hat{u}e^{-i(k(n+2)\Delta x-v(\Delta t))}] = Re[\hat{u}e^{-i(nk\Delta x+2k\Delta x-v(\Delta t))}] \\ &= Re[\hat{u}e^{-i(2k\Delta x)}e^{-i(nk\Delta x-v\Delta t)}] \end{aligned}$$

$$u(j = n + 2, m = 1) = Re[\hat{u}(e^{-i(2k\Delta x)}e^{-i(nk\Delta x-v\Delta t)})]$$

$$\begin{aligned} u(j = n - 2, m = 1) &= Re[\hat{u}e^{-i(k(n-2)\Delta x-v(\Delta t))}] = Re[\hat{u}e^{-i(nk\Delta x-2k\Delta x-v(\Delta t))}] \\ &= Re[\hat{u}e^{i(2k\Delta x)}e^{-i(nk\Delta x-v\Delta t)}] \end{aligned}$$

$$u(j = n + 2, m = 1) = Re[\hat{u}(e^{i(2k\Delta x)}e^{-i(nk\Delta x-v\Delta t)})]$$

$$\begin{aligned}
& -(v)^2 \operatorname{Re}[\hat{u}(e^{-i(nk\Delta x - v\Delta t)})] \\
&= \frac{gH}{4d'^2} \left(\left(\operatorname{Re}[\hat{u}(e^{-i(2k\Delta x)} e^{-i(nk\Delta x - v\Delta t)})] + \operatorname{Re}[\hat{u}(e^{i(2k\Delta x)} e^{-i(nk\Delta x - v\Delta t)})] \right) \right) \\
& - \left(2 \operatorname{Re}[\hat{u} e^{-i(kn\Delta x - v(\Delta t))}] \right) - f^2 \operatorname{Re}[\hat{u} e^{-i(kn\Delta x - v(\Delta t))}]
\end{aligned}$$

$$-(v)^2 = \frac{gH}{4d'^2} \left(\left([(e^{-i(2k\Delta x)})] + [(e^{i(2k\Delta x)})] \right) - (2) \right) - f^2$$

$$(v)^2 = f^2 - \frac{gH}{4d'^2} \left(\left([(e^{-i(2k\Delta x)})] + [(e^{i(2k\Delta x)})] \right) - (2) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{4f^2 d'^2} \left(\left([(e^{i(2k\Delta x)} + e^{-i(2k\Delta x)})] \right) - (2) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{2f^2 d'^2} \left(\left(\left(\frac{e^{i(2k\Delta x)} + e^{-i(2k\Delta x)}}{2} \right) - 1 \right) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{2f^2 d'^2} ((\cos(2k\Delta x) - 1))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{2f^2 d'^2} (1 - \cos(2k\Delta x))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{f^2 d'^2} \frac{1}{2} (1 - \cos(2k\Delta x))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{f^2 d'^2} \sin^2(k\Delta x)$$

$\cos x = \operatorname{Re}(e^{-ix}) = \frac{e^{ix} + e^{-ix}}{2}$	$\sin x = \operatorname{Im}(e^{-ix}) = \frac{e^{ix} - e^{-ix}}{2i}$
$\sin^2 x + \cos^2 x = 1$	$\sin^2 x = \frac{1}{2} (1 - \cos(2x))$

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